

REPORT

PROBLEM STATEMENT

Financial institutions face major financial losses due to loan defaults caused by inaccurate credit risk assessment. Traditional credit evaluation methods are often time-consuming and may fail to identify high-risk applicants effectively. This project aims to develop a machine learning-based Credit Risk Modeling system that predicts the likelihood of loan default based on customer financial and personal information. The system helps banks and financial institutions make faster, more accurate, and data-driven lending decisions, reducing financial risk and improving loan approval efficiency.

Objective

The main objective of this project is to build an intelligent credit risk analysis system that can classify loan applicants into risk categories and support better decision-making for loan approvals.

Specific Objectives

To analyze customer financial and personal data to identify important factors affecting credit risk.

To preprocess and clean the dataset for accurate model training and prediction.

To build and train machine learning models for predicting loan default probability.

To classify applicants into Low Risk, Medium Risk, and High Risk categories.

To generate recommendations such as Approve Loan, Review Application, or Reject Loan based on prediction results.

To create an interactive frontend using Streamlit for easy user input, prediction, and report generation.

To provide downloadable reports for business use and decision-making support.

Conclusion

The Credit Risk Modeling project provides a practical solution for predicting loan default risk using machine learning techniques. By analyzing customer data and identifying risk levels, the system helps financial institutions make smarter lending decisions and reduce potential financial losses. The integration of frontend visualization, risk categorization, recommendation generation, and downloadable reports makes the project business-oriented and industry-

REPORT

relevant. This project demonstrates both technical implementation and real-world problem-solving capability in the finance and data analytics domain.